ASE Advanced Packaging Solutions

Yin Chang
SVP / Marketing

aseglobal.com

September 1st 2023

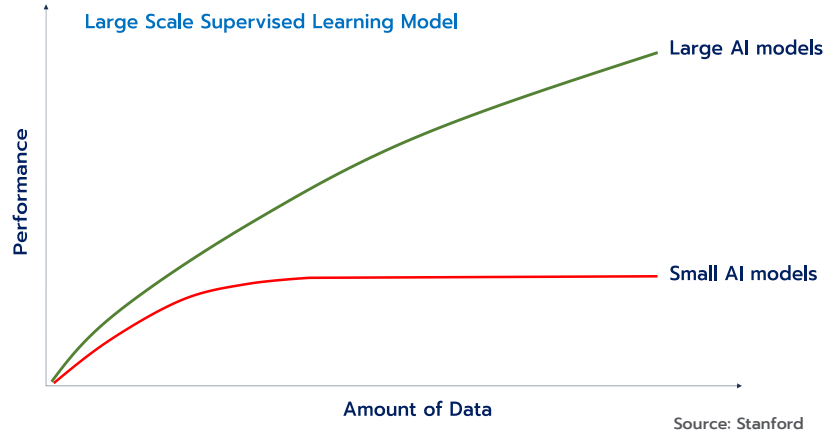
The semiconductor industry landscape:

>$1T

Growing almost double by 2030

$600M
2022

$205M
2000



Source: Stanford

PC Era | Smartphone Era | Data Center Era | AI Era

2000 2005 2010 2015 2020 2025 2030

Source: Gartner 2023

Our world requires massive scaling of compute processing capabilities – and it all starts with semiconductor innovation






AI everywhere







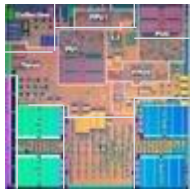


Technology convergence

- SiP—System integrated on board (off package) have driven by OSATs/EMS in semiconductor markets
- Si CMOS (foundry-focused) to CSYS (Complementary Systems) (adv. packaging-focused) —a new era for packaging
- **Multi-die & Chiplets** heterogenous integration on substrate (on package) will become the mainstream in the future



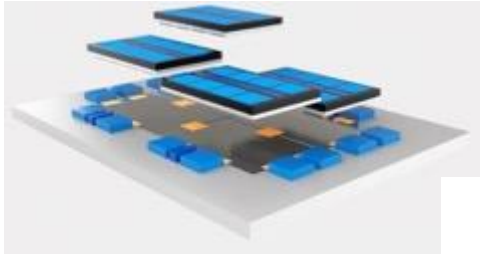
1990s
SoC



2000s
SoC w/ IP



2010s
MCM



2020s

CSYS
Multi-die
Chiplets
IDMs/Foundry/
OSTAs/EMS
Ecosystem



2010S
SiP



2000s
HDI PCB



1990s
FR4 PCB

IDM or Foundry

Foundry & 3rd Party IP

IDM, Foundry & OSAT

OSAT

EMS

EMS

Scaling for the AI era

MORE THAN MOORE

Market demand for AI performance is faster than Moore's Law doubling transistors every 18 months

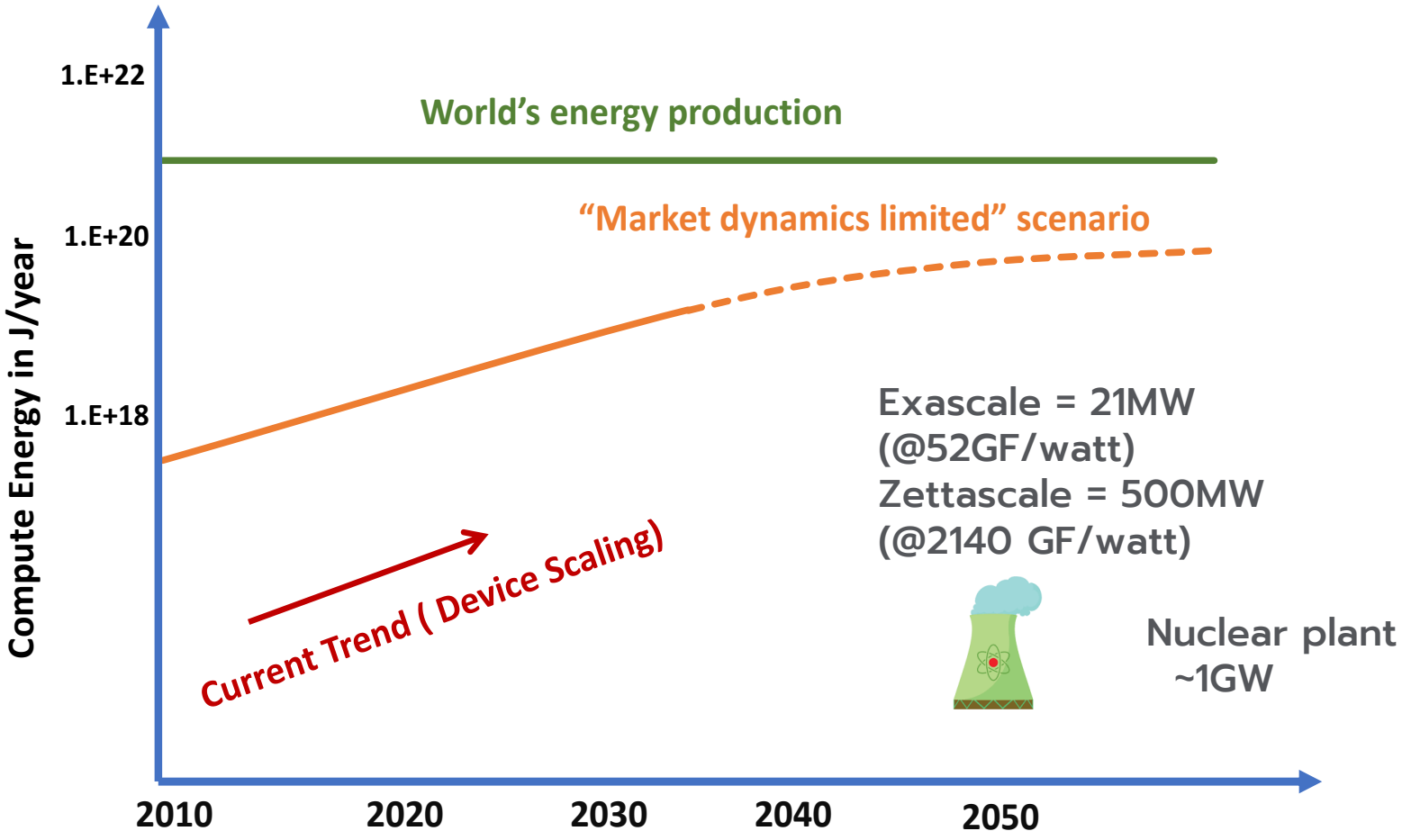
- AI/ML performance has increased nearly 6.8x~11x in the past two years from 2021~ 2022 that outstrips and more than Moore's Law.



Scaling for the AI era

PUSHING LIMITS

An inflection point is approaching that requires unprecedented innovation and collaboration across the value chain

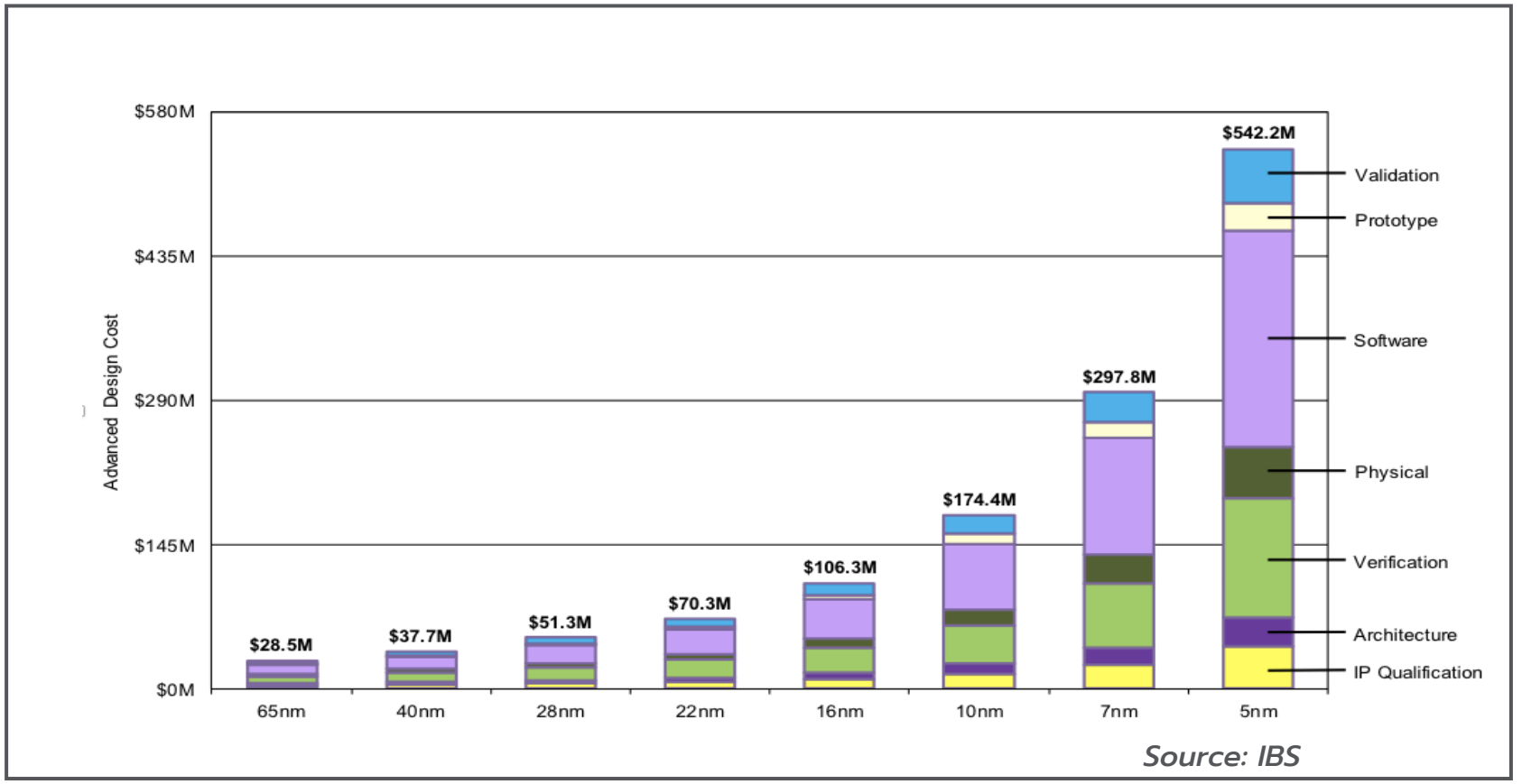


Scaling for the AI era

LOWERING COSTS

New node introduction rate is slowing, yet cost continues to rise requiring novel design, process, and manufacturing techniques

Si Node Dev Cost



Source: IBS

Scaling for the AI era

MORE THAN MOORE
Market demand for AI performance is faster than Moore's Law doubling transistors every 18 months

PUSHING LIMITS
An inflection point is approaching that requires unprecedented innovation and collaboration across the value chain

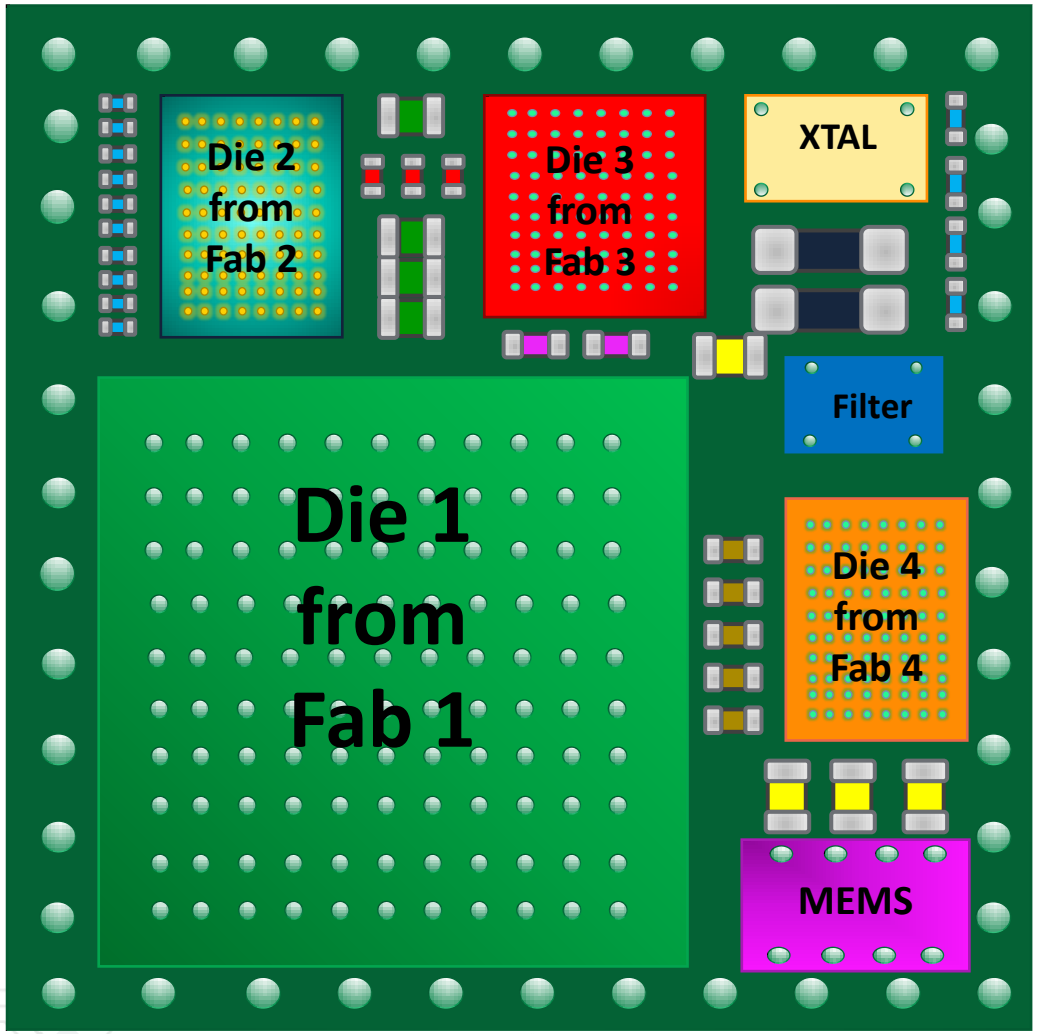
LOWERING COSTS
New node introduction rate is slowing, yet cost continues to rise requiring novel design, process, and manufacturing techniques

NEW EFFICIENCIES
Chips need more functional blocks: chip size expanding each generation while optimizing new architecture, packaging, and integration technologies

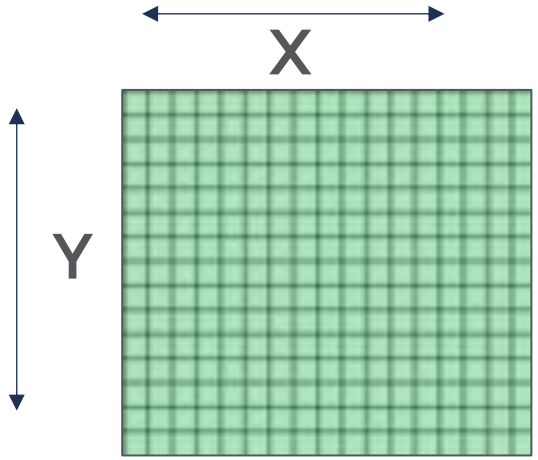
High Density Heterogeneous integration using advanced packaging is a key enabler

Introducing Heterogeneous Integration:

Heterogeneous by material, component type, circuit type, node and bonding/interconnect method & sources

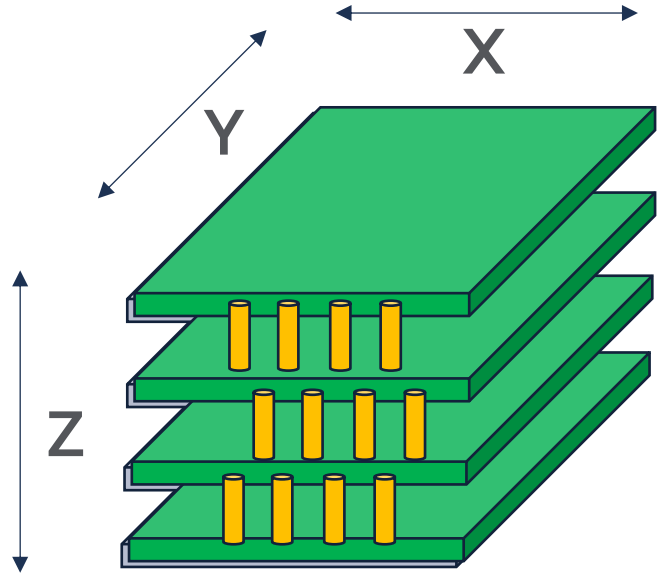


Evolution of Integration Platforms:



Moore's Law

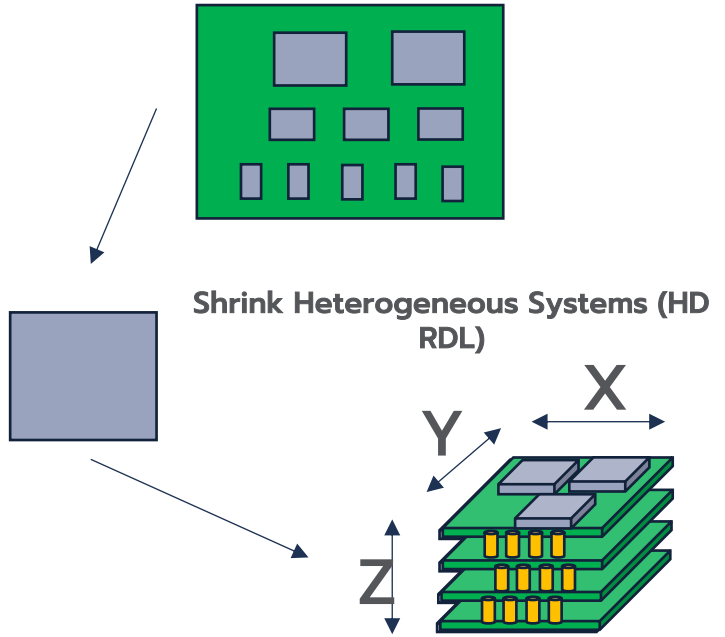
2D



3D Packaging (Monolithic Chips)

3D

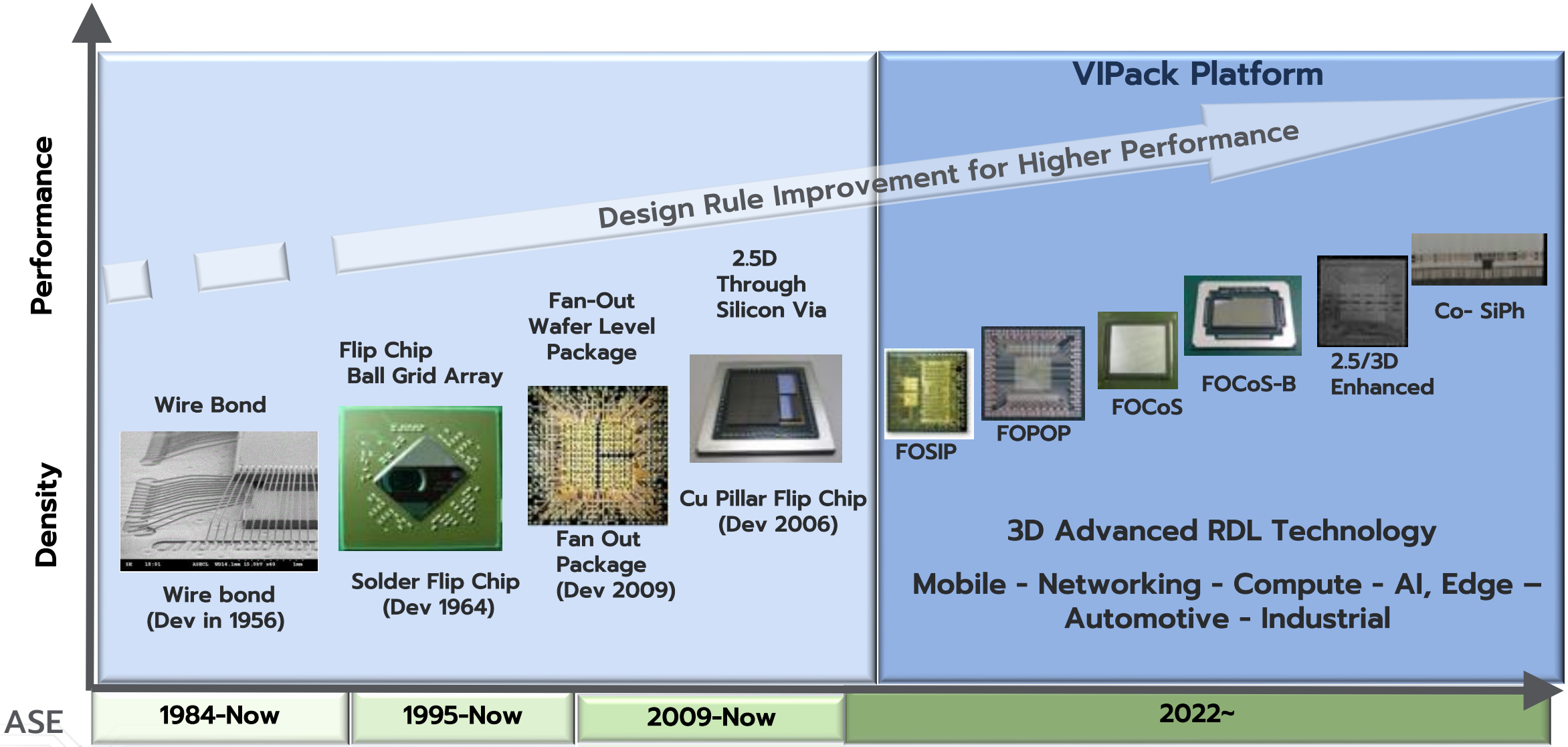
Advanced Packaging



Stacking Heterogeneous Systems (Open Ecosystem)

VIPack™

ASE Advanced Packaging Roadmap



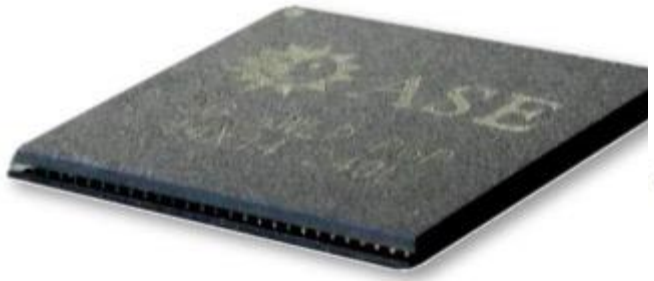
VIPack™: Six Technology Pillars

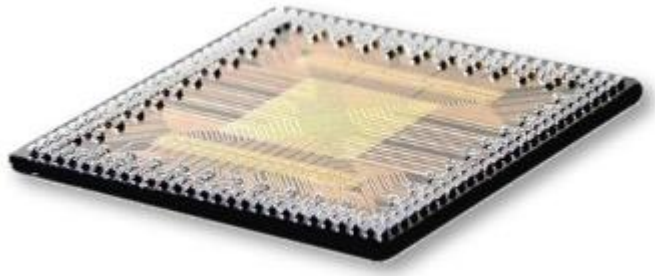


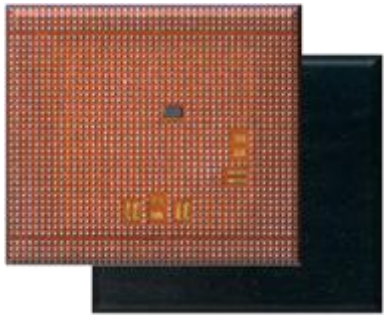
I	II	III	IV	V	VI
FOPoP	FOCoS	FOCoS – Bridge (Embedded)	FOSiP	2.5D/3D IC	Co-Packaged Optics
Integrated Design Ecosystem					

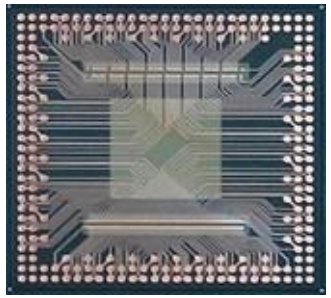
VIPack™ Pillar I: FOPoP









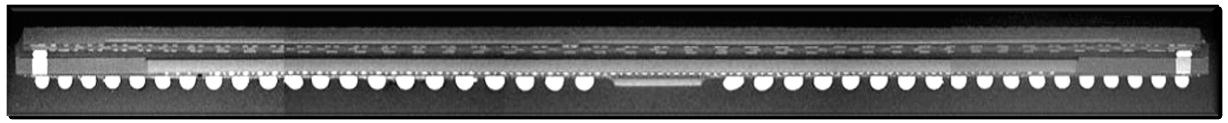


Key Features:

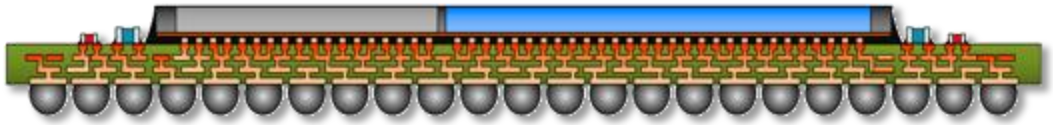
- FOPOP: Fan Out Package-on-Package
- RDL Based Package with 2 Routing Planes and Fine Pitch Cu Posts Between Layers
- Bottom Package Height <330um
- Land Side Cap or Embedded DTC (M)
- Cu Post Pitch ~140um (@100um Die)
- Power Delivery Solution with Near Die DTC (Adv. Silicon Node)

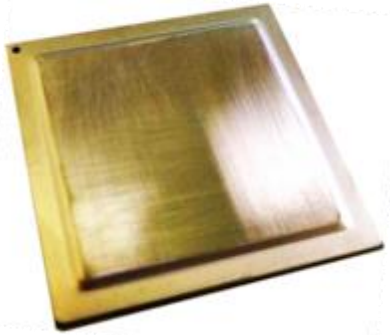
Applications:

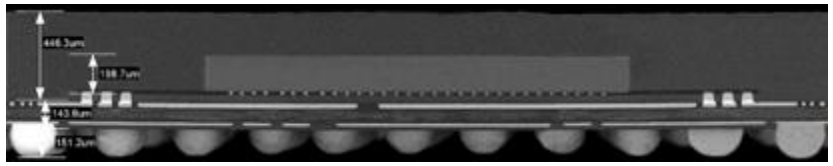
- Application Processor (AP) Mobile
- Co-Package SiPh
- Mobile/Auto AIP (Antenna-In-Package)



VIPack™ Pillar II: FOCoS



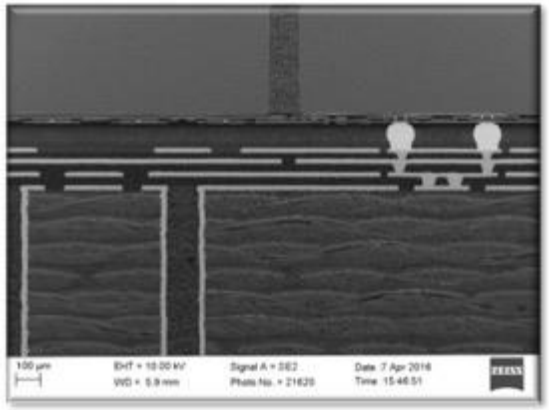




Key Features:

- High Density Fan-Out Single or Multi Die Integration Package-Chip First or Chip Last
- 1.5/1.5um Line/Space, 6 metal layers, 32x38mm FO
- Flip Chip attached to RDL or Organic Substrate
- Stacked Via for Low Impedance
- Support HBM2/3 Integration

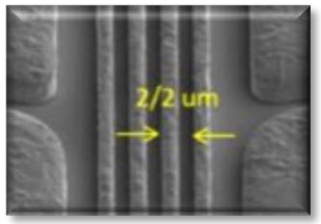


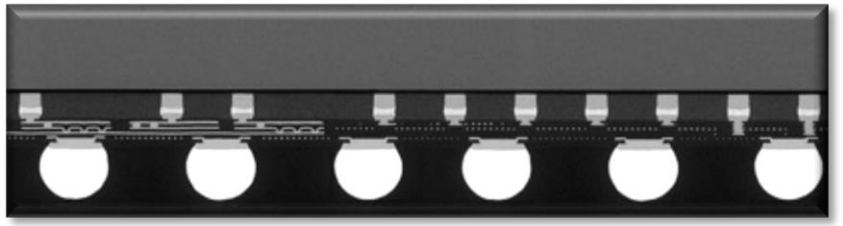


Applications:

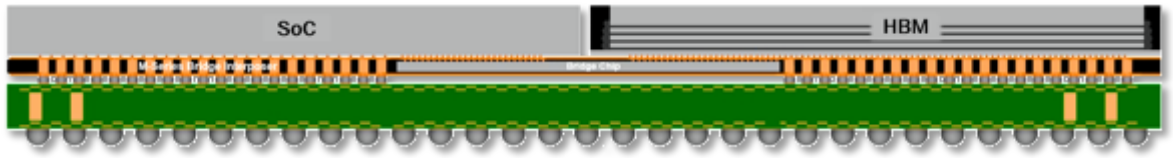
- APU + Memory
- GPU + Memory
- Networking
- SiP/Modules
- AI

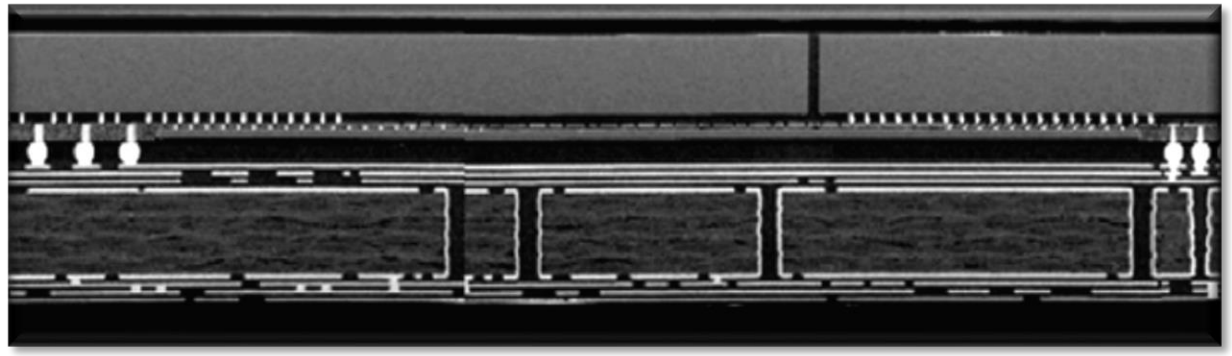






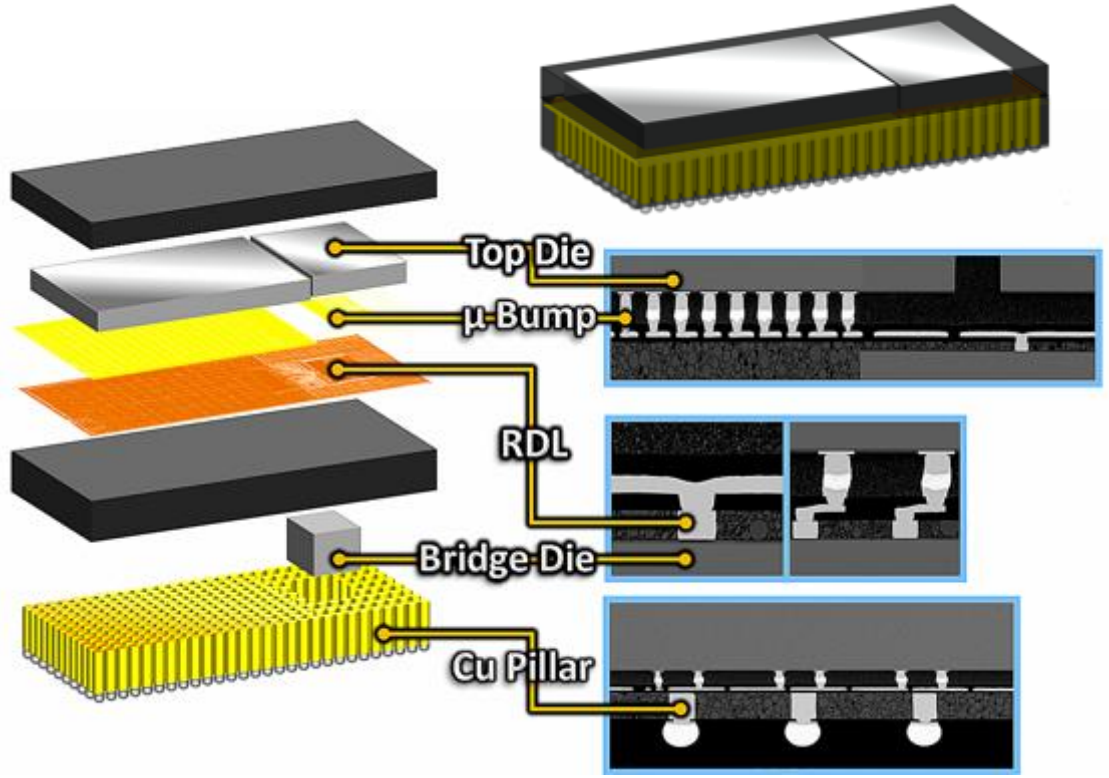
VIPack™ Pillar III: FOCoS Bridge (Embedded)





Key Features:

- Embedded Die Layer- Die Attached on Both Sides of RDL (Bridge, DTC, Power Mgmt, Active)
- High Density Fan-Out Single or Multi Die Integration Package-Chip First or Chip Last
- 1.5/1.5um Line/Space, 6 metal layers, 32x38mm FO
- Flip Chip attached to RDL or Organic Substrate
- Lower Cross Talk (No TSV)



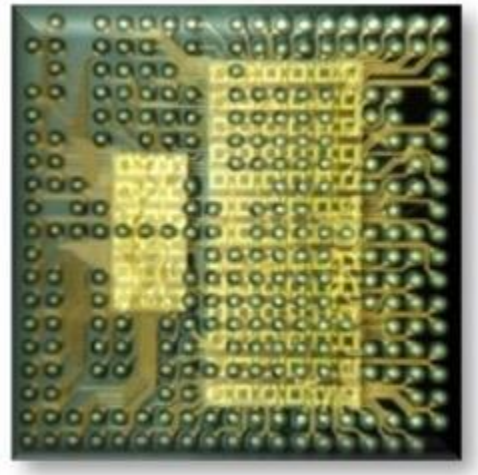
Applications:

- APU + Memory
- GPU + Memory
- Switching/Router-Networking/AI
- SiP/Modules

VIPack™ Pillar IV: FOSiP



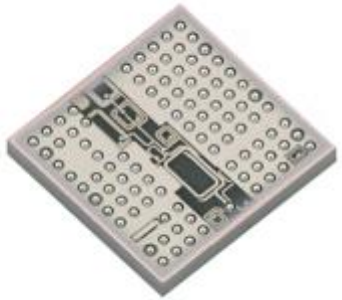


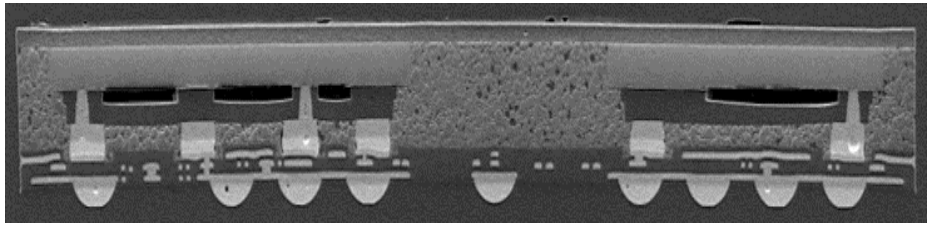


Key Features:

- FOSIP: Fan-Out System in Package
- Lowest Profile Structure
- Ultra High-Density Component spacing- 50um
- Multi Layer RDL Substrate Layer
- Integration of both un-packaged and packaged components
- The option for double sided RDL – Chip Last for Std Component Termination



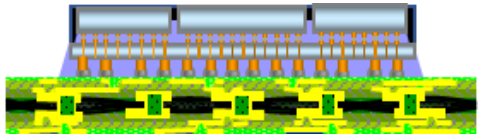


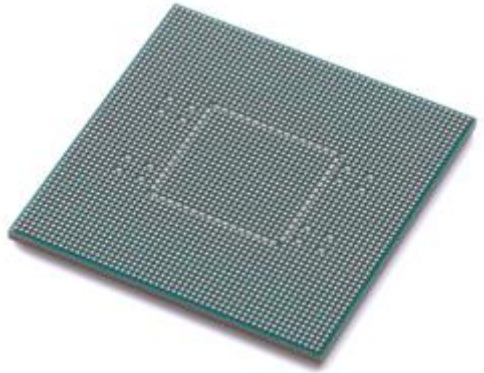


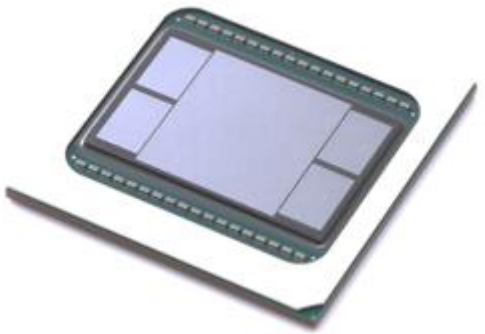
Applications:

- Mobile - RFFEM
- IoT- WIFI
- Auto - Radar

VIPack™ Pillar V: 2.5D/3D TSV







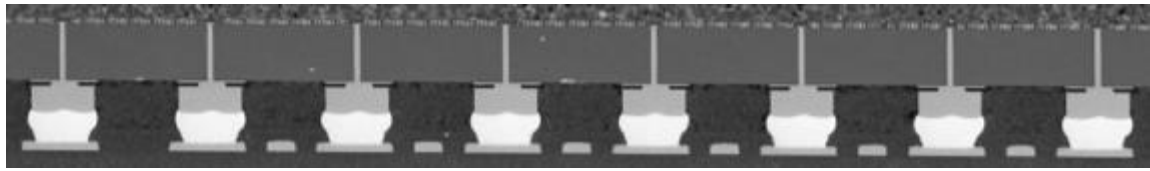
Key Features:

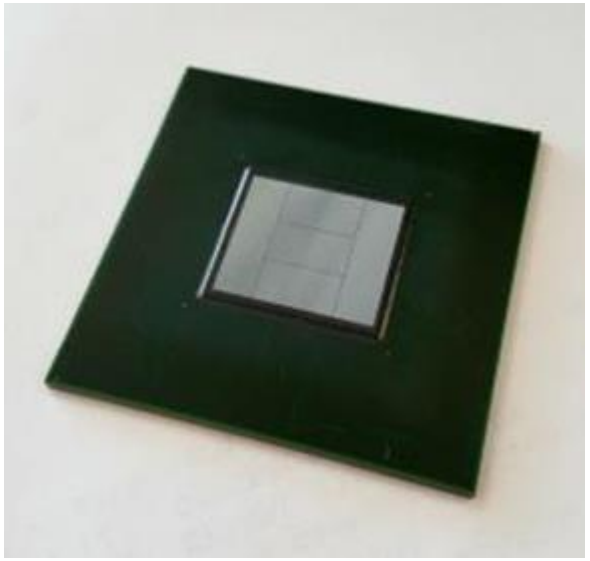
2.5D Chiplet Integration

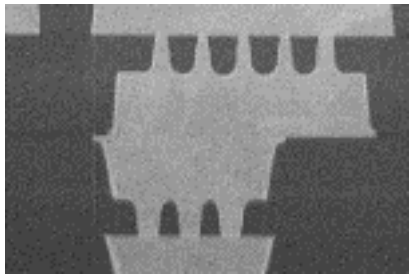
- **Stitched interposer > 1.6x reticle**
- Interposer BEOL RDL Passive Integration Options
- Fine Pitch D2D Interconnects: μBump pitch ≤ 45μm, L/S ≤ 1μm, up to 5 layers

3D Die Stacking Solutions

- **Active interposer: DRAM Interposer -** high memory bandwidth & energy efficiency
- 3D die stacking: DRAM die stacking - high memory bandwidth & capacity
- **Ultra Fine Pitch Interconnects: Bond pitch < 30μm**



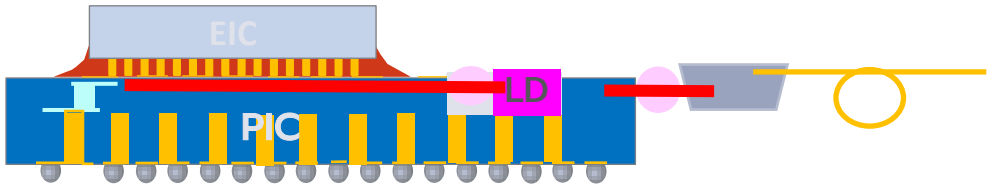


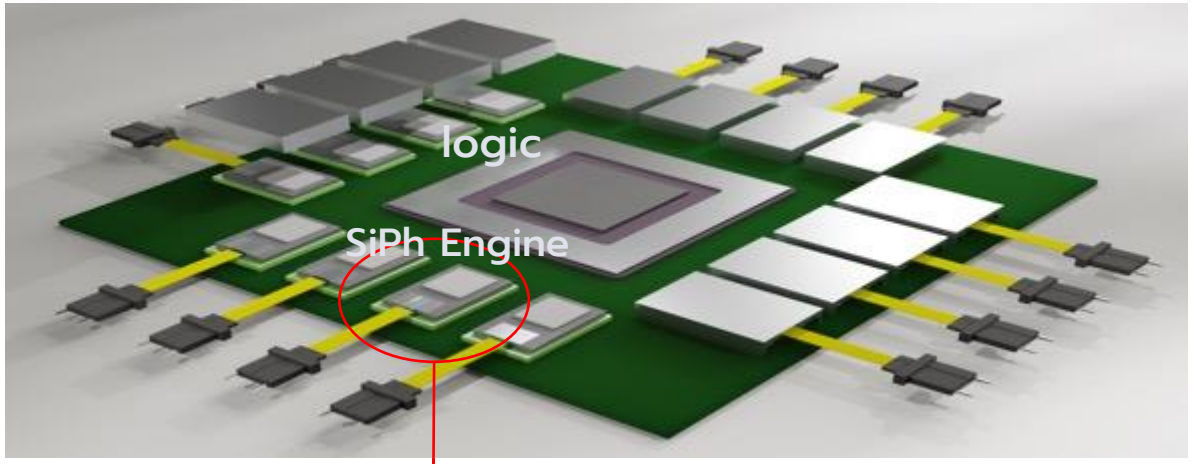


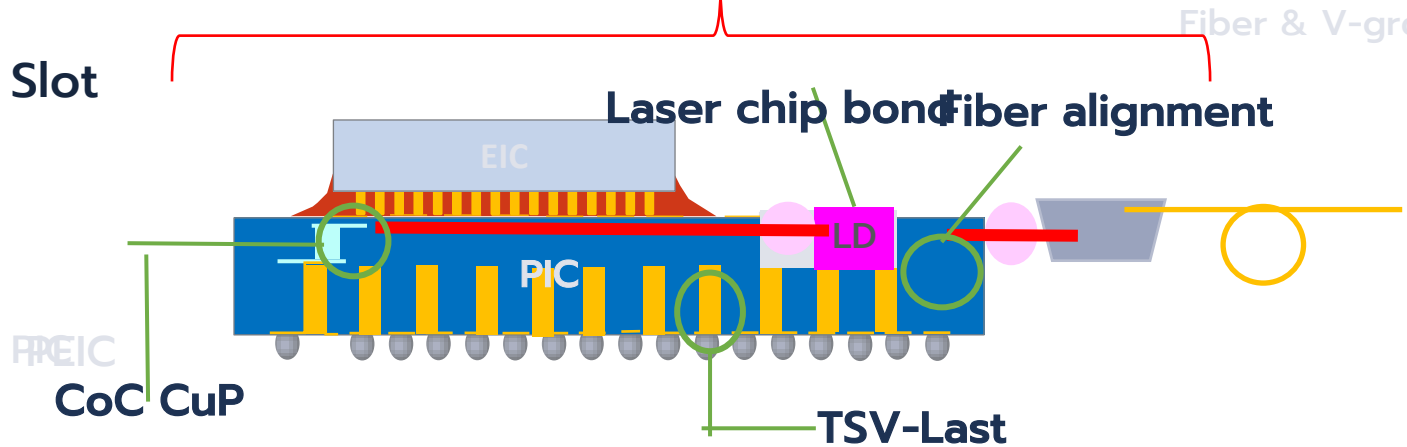
Applications:

- **High Power Computing, Networking, Mobile**

VIPack™ Pillar VI: Integrated Optics







Key Features:

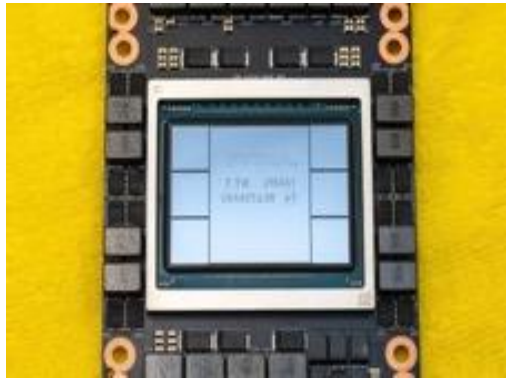
- Replaces Traditional SERDES HS Interface
- RDL Based Package with 2 Routing Planes and Fine Pitch Cu Posts Between Layers
- MicroBump- 45um pitch, 30um Diam
- 2 RDL Layers
- TSV Last Pitch 60um dia, 140um pitch- Slot design version 30um dia, 60um pitch
- V-Groove for Alignment- DRIE Process
- Power Delivery Solution with Near Die DTC (Adv. Silicon Node)

Applications:

- ASIC on Network Switch
- Stand Alone Laser Engine- For High Speed

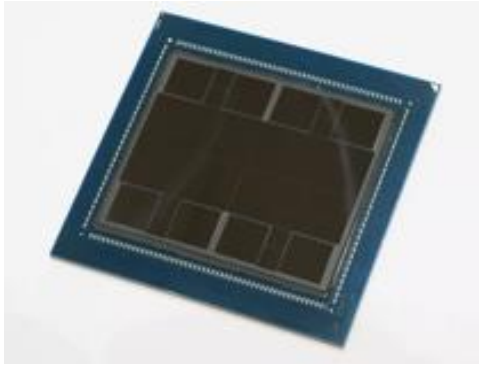
Advanced Packaging for AI / HPC

- Successful AI/HPC Hardware Deployment Requirement:
  - Multiple chiplets, components heterogenous integration
  - High density interconnect, high bandwidth, low latency and high-power efficiency and low cost (PPAC)
  - Large Body size (Interposer, Substrate, HBM2e/HBM3)
- 2.5D Si TSV Technology Trend
  - Large Si TSV interposer size ~ 3.3X with 8 HBM3
  - RDL interposer (bridge solution)



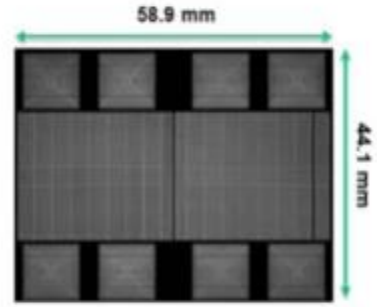
GPU + 6HBM3





APU + 8HBM3

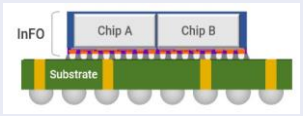
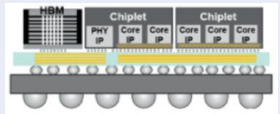
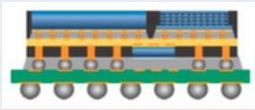
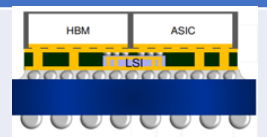




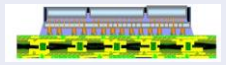
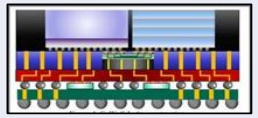
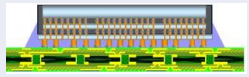
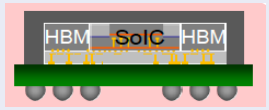
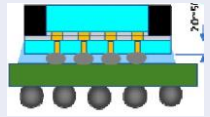
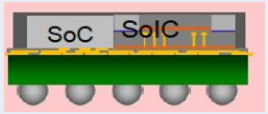
5th Gen CoWoS (2) SoCs, a (1) Chiplet and eight (8) HBM

ASE Advanced Technology Portfolio Readiness

ASE Advanced FOCoS Solutions Readiness

Technology	FO Chip First		FO Chip Last		FO Bridge	
Comparison	**ASE FOCoS-CF**	**TSMC InFO-R/So-W**	**ASE FOCoS-CL**	**TSMC CoWoS-R**	**ASE FOCoS-B**	**TSMC InFO-L/CoWoS-L**
Structure						
Status	HVM (2016)	HVM (2018)	Qualified	Qualified	HVM (2022)	HVM (2021)

ASE Advanced 2.5D & 3D readiness

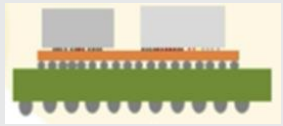
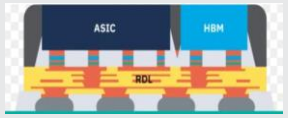
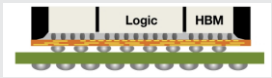
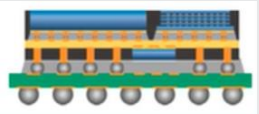
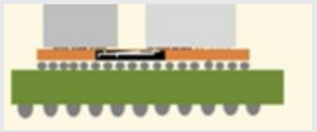
Technology	2.5D Si TSV		3D			
Comparison	**ASE 2.5D Si TSV**	**TSMC CoWoS-S**	**ASE 3D - ubump**	**TSMC SoIC-P (ubum)**	**ASE 3D (Cu-Cu) Hybrid Bonding**	**TSMC SoIC-X (Cu-Cu) Hybrid Bonding**
Structure						
Status	HVM	HVM	Qualified	Qualified	Development	HVM

ASE Advanced Packaging Technology

- Advanced packaging solutions offer design creativity to enable Si system performance and efficiency that meet the needs of AI and HPC
- New proficiencies in the semiconductor design (chiplets) and manufacturing process support integration technologies (Si and organic interposer) being pioneered to deliver next level computation advances
- ASE VIPack™ is an advanced packaging platform designed to enable vertically integrated package solutions and improve efficiency, speed time-to-market, and sustain profitable growth.
- VIPack™ evolving through existing and advanced RDL package technologies to achieve:
  - Maximized clock speed and performance through best-in-class silicon providers
  - Optimized co-design time, development and product time to market
  - Open silicon ecosystem in partnership with our foundries/supply chain

Thank you

ASE Advanced Technology Portfolio Readiness

Technology	FO Chip First		FO Chip Last		FO Bridge	
Comparison	ASE FOCoS-CF	Samsung I-Cube CF	ASE FOCoS-CL	Samsung I-Cube CL	ASE FOCoS-B	Samsung I-CubeE (Embedded Bridge)
Structure						
Status	HVM (2016)	Qualified	Qualified	Qualified	HVM (2022)	Under Qual